

AP EAMCET
Engineering Entrance Exam
24th April 2018- Shift 2

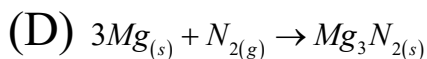
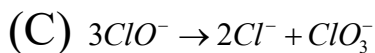
Chemistry

1. If the kinetic energy of an electron of mass 9.0×10^{-31} kg is 8.0×10^{-25} J. the wavelength of this electron in nm is
A. 104.1
B. 276.2
C. 552.2
D. 828.4
2. The radius ratio of Bohr's first orbit of hydrogen like species He^+ , Li^{2+} and Be^{3+}
A. 3:4:6
B. 2:3:4
C. 6:4:3
D. 4:3:2
3. The element Uuq contains 12.8% more neutrons compared to protons. What is the approximate mass number of it?
A. 263
B. 253
C. 233
D. 243
4. Observe the following statements

- i. The dipole moment of NH_3 is higher than the dipole moment of NF_3 .
- ii. The dipole moment of chloroform is zero.
- iii. Covalent bond character in NaCl is more compared to CuCl .

The statements which are not correct are

- A. I, iii
 - B. II, iii
 - C. i, II, iii
 - D. I, ii
5. The correct order of energy of σ_{2p} , π_{2p} , π_{2p}^* and σ_{2p}^* orbitals in F_2 is
- A. $\sigma_{2p} > \pi_{2p} > \pi_{2p}^* > \sigma_{2p}^*$
 - B. $\sigma_{2p} < \pi_{2p} < \pi_{2p}^* < \sigma_{2p}^*$
 - C. $\pi_{2p} > \sigma_{2p} > \pi_{2p}^* > \sigma_{2p}^*$
 - D. $\pi_{2p} < \sigma_{2p} < \pi_{2p}^* < \sigma_{2p}^*$
6. If the most probable speed of methane at a certain temperature is 400 ms^{-1} , the kinetic energy of one mole of methane at the same temperature in J is
- A. 1024
 - B. 2048
 - C. 3072
 - D. 1920
7. Which of the following are disproportionation reactions ?
- (A) $2\text{NO}_{2(g)} + 2\text{NaOH}_{(ag)} \rightarrow \text{NaNO}_{2(ag)} + \text{NaNO}_{3(ag)} + \text{H}_2\text{O}_{(l)}$
 - (B) $\text{Cl}_{2(g)} + 2\text{NaOH}_{(ag)} \rightarrow \text{NaClO}_{(ag)} + \text{NaCl}_{(ag)} + \text{H}_2\text{O}_{(l)}$



A.A,B,D

B.B,C,D

C.B,C

D.A,B,C

8. If the work done during the isothermal reversible expansion of an ideal gas at a pressure of 10 atm from 4L to a final volume is -184.24 L atm . the final volume of the gas in L is
- A. 40
B. 400
C. 8
D. 80
9. The equilibrium concentrations of N_2 , H_2 and NH_3 in the formation of NH_3 at 500 K are $1.25 \times 10^{-2} \text{ M}$, $4.0 \times 10^{-2} \text{ M}$ and $1.6 \times 10^{-2} \text{ M}$ respectively. The equilibrium constant K_p at the same temperature is
- A. $3.2(RT)^{-2}$
B. $32(RT)^2$
C. $320(RT)^2$
D. $320(RT)^{-2}$
10. If the pH of 0.10 M monoacidic base at 298 K is 9.0. the value of K_b and $\text{p}K_b$ at the same temperature are respectively
- A. 1.0×10^{-9} , 9.0
B. 1.0×10^{-5} , 5.0

C. 1.0×10^{-10} , 10.0

D. 1.0×10^{-4} , 4.0

11. 11. The correct statements from the following are :

- i. $\text{Ca}(\text{OH})_2$ is used in Clark's method of removal of temporary hardness of water
- ii. 100 mL of 10 vol H_2O_2 . on complete decomposition liberates 1L of oxygen at STP.
- iii. Urea can be added to the solution of H_2O_2 as stabilizer.

A.i, iii

B.ii, iii

C.i, iii

D.i, ii, iii

12. Identify the answer in which. the trends corresponding to hydroxides of group 2 elements (from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$) are correctly shown
basic character chemical stability solubility decreases

A.increases increases decreases

B.decreases decreases decreases

C.decreases increases increases

D.increases increases increases

13. in which of the following reaction, H_2 is liberated?

i. $\text{Al}(s) + \text{HCl}(aq) \rightarrow$

ii. $\text{Al}(s) + \text{NaOH}(aq) \rightarrow$

iii. $\text{B}_2\text{H}_6(g) + \text{H}_2\text{O}(l) \rightarrow$

iv. $\text{F}_2(g) + \text{H}_2\text{O}(l) \rightarrow$

A.i, ii, iii, iv

B.i, ii, iv

C.i, ii, iii

D.ii, iii, iv

14. Identify the element whose monoxide and dioxide are acidic in nature.

A.Sn

B.C

C.Ge

D.Pb

15. Methemoglobinemia is caused by the presence of X in drinking water. X is

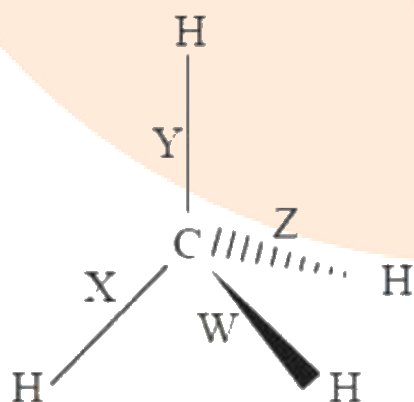
A.1000 ppm of sulphate

B.20 ppm of fluoride

C.200 ppm of nitrate

D.50 ppb of lead

16. The three dimensional structure of CH_4 is given below. The bonds are labelled as W, X, Y and Z. What are the bonds lying in the plane (A), projecting out of the plane away from the observer (B) and projecting out of the plane towards the observer (C) ?



A B C
A.X,Y Z W

A B C
B.Z w X,Y

A B C
C.X,Y W Z

A B C
D.W X,Y Z

17. Assertion (A) : Reaction of 1-butene with HBr gives 1-bromobutane as major prgermm Reason (R) : Addition of HBr to unsymmetrical alkenes proceeds according to Markovnikov's rule

The correct answer is

A.(A) and (R) are correct. (R) is the correct explanation of (A)

B.(A) and (R) are correct but (R) is not the correct explanation of (A)

C.(A) is correct but (R) is not correct

D.(A) is not correct but (R) is correct

18. Which one of the following. is not aromatic ?

A.Cyclopentadienyl anion

B.Benzene

C.Cyclopentadienyl cation

D.Cyclopentadienyl cation

19. If a metal crystallises in a face centred cubic structure with metallic radius of 25 pm. the number of unit cells in 1.0 cm^3 of lattice is

A. 2.282×10^{28}

B. 1.414×10^{28}

C. 1.414×10^{24}

D. 2.828×10^{24}

20. Match the following.

List - I

A) Azeotrope

B) Henry's law

C) Cryoscopic constant

D) Van't Hoff factor

List - II

$$\Delta T_b = iK_b m$$

$$p = K_H x$$

$$\Delta T_f / m$$

Deviation from Raoult's law

$$\pi = CRT$$

	(A)	(B)	(C)	(D)
A.	II	III	V	IV
	(A)	(B)	(C)	(D)
B.	IV	II	III	I
	(A)	(B)	(C)	(D)
C.	IV	II	I	III
	(A)	(B)	(C)	(D)
D.	I	III	II	IV

21. The mass of ascorbic acid ($\text{C}_6\text{H}_8\text{O}_6$) to be dissolved in 100 g of acetic acid to lower its freezing point by 1.5°C is : (K_f for acetic acid is $4.0 \text{ K kg mol}^{-1}$)

A. 17.6

B. 8.8

C.6.6

D.13.2

22. At T(K). the molar conductivity of 0.04 M acetic acid is $7.8 \text{ S cm}^2 \text{ mol}^{-1}$. If the limiting molar conductivities of H^+ and CH_3COO^- at T(K) are 349 and $41 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. the dissociation constant of acetic acid is

A. 1.63×10^{-5}

B. 8.33×10^{-5}

C. 1.63×10^{-4}

D. 8.33×10^{-4}

23. At T(K). if the rate constant of a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. the time to reduce the initial concentration of the reactant to 1/10 in seconds is :

A.1500

B.1000

C.100

D.50

24. Which one of the following statements is correct ?

A.The difference in the initial and final concentrations of the adsorbate gives the value of 'x' in Freundlich equation ($x = \text{mass of adsorbate}$)

B.The mass of adsorbent gives the value of 'n' in Freundlich equation

C.Chemisorption decreases with increase of surface area of the adsorbent

D.Enthalpy of adsorption is 20 kJ mol^{-1} in chemisorption

25. Blister copper is extracted from copper pyrites. After removal of slag FeSiO_3 . blister copper is obtained by

- A.self reduction
B.reduction with CO
C.reduction with coke
D.self oxidation
26. Which one of the following is used to remove moisture from ammonia gas ?
A. P_4O_{10}
B. Anhydrous $CaCl_2$
C. Quick lime
D. Conc H_2SO_4
27. In the reaction $2I^- + H_2O(l) + O_3(g) \rightarrow 2OH^-(aq) + I_2(s) + O_2(g)$ ozone acts as
A. an oxidizing agent
B. a reducing agent
C. a dehydrating agent
D. a hydrating reagent
28. X_2 reacts with water to form $HX(aq)$ and $HOX(aq)$. Y_2 reacts with water to form $O_2(g)$. $Y^-(aq)$ and $H^+(aq)$. Z_2 does not react with water. X_2, Y_2, Z_2 are respectively
A. I_2, Cl_2, F_2
B. Cl_2, F_2, I_2
C. F_2, Cl_2, I_2
D. Cl_2, Br_2, I_2
29. Identify the statements which are not correct.
i. CrO_3 and Mn_2O_7 are basic oxides
ii. V_2O_3 and V_2O_4 are acidic oxides
iii. $Cr(VI)$ is more stable than $W(VI)$

- A.I, ii
- B.Ii, iii
- C.i, iii
- D.i, ii, iii

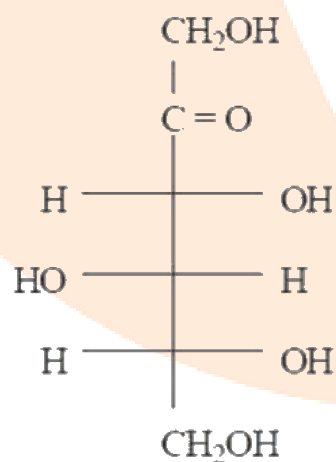
30. Identify the complex which exhibits optical isomerism.

- A. $\text{Co}(\text{NH}_3)_5 (\text{SO}_4) \text{Br}$
- B. $\text{Co}(\text{NH}_3)_5 \text{Cl}_3$
- C. $\text{Co}(\text{en})_3 \text{Cl}_3$
- D. $\text{Co}(\text{NH}_3)_4 (\text{NO}_2) \text{Cl}$

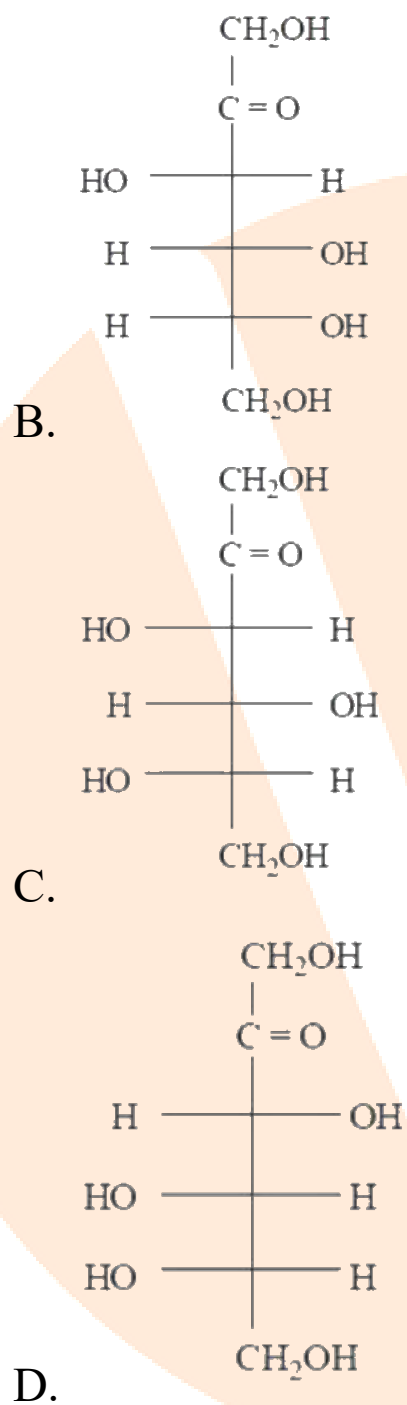
31. Examples of thermoplastic polymer (X) and thermosetting polymer (Y) are

- | X | Y |
|----------------|-----------|
| A. teflon | neoprene |
| B. neoprene | glyptal |
| C. polystyrene | bakelite |
| D. bakelite | polythene |

32. Fischer projection formula of D – (-) – fructose is



A.



33. The drug used for relieving tension is

A. Teifenadine (Seldane)

B. Alitame

C. Meprobamate

D. Aspartame

34. Chloroethane with silver acetate forms X and with LiAlH_4 forms Y. What are X and Y?

X	Y
$\text{H}_3\text{CCH}_2\text{CH}_3$	H_3CCHO
$\text{HCOOCH}_2\text{CH}_3$	$\text{H}_3\text{CCH}_2\text{OH}$
$\text{H}_3\text{CCOOCH}_2\text{CH}_3$	H_3CCH_3
$\text{H}_3\text{CCH}_2\text{COOH}$	$\text{H}_3\text{CCH}_2\text{Li}$

35. Pent-1-ene reacts with diborane to form X. X on oxidation using H_2O_2 in the presence of aqueous NaOH gives Y.

Compound Y is

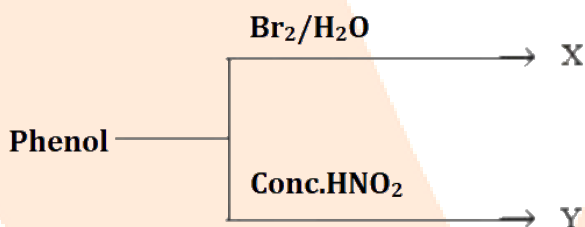
A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

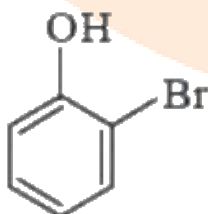
C. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$

D. $\text{CH}_3\text{CH}_2\text{C}(\text{OH})(\text{CH}_3)\text{CH}_3$

36. Identify X and Y in the following reactions

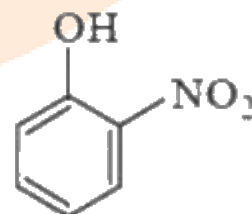


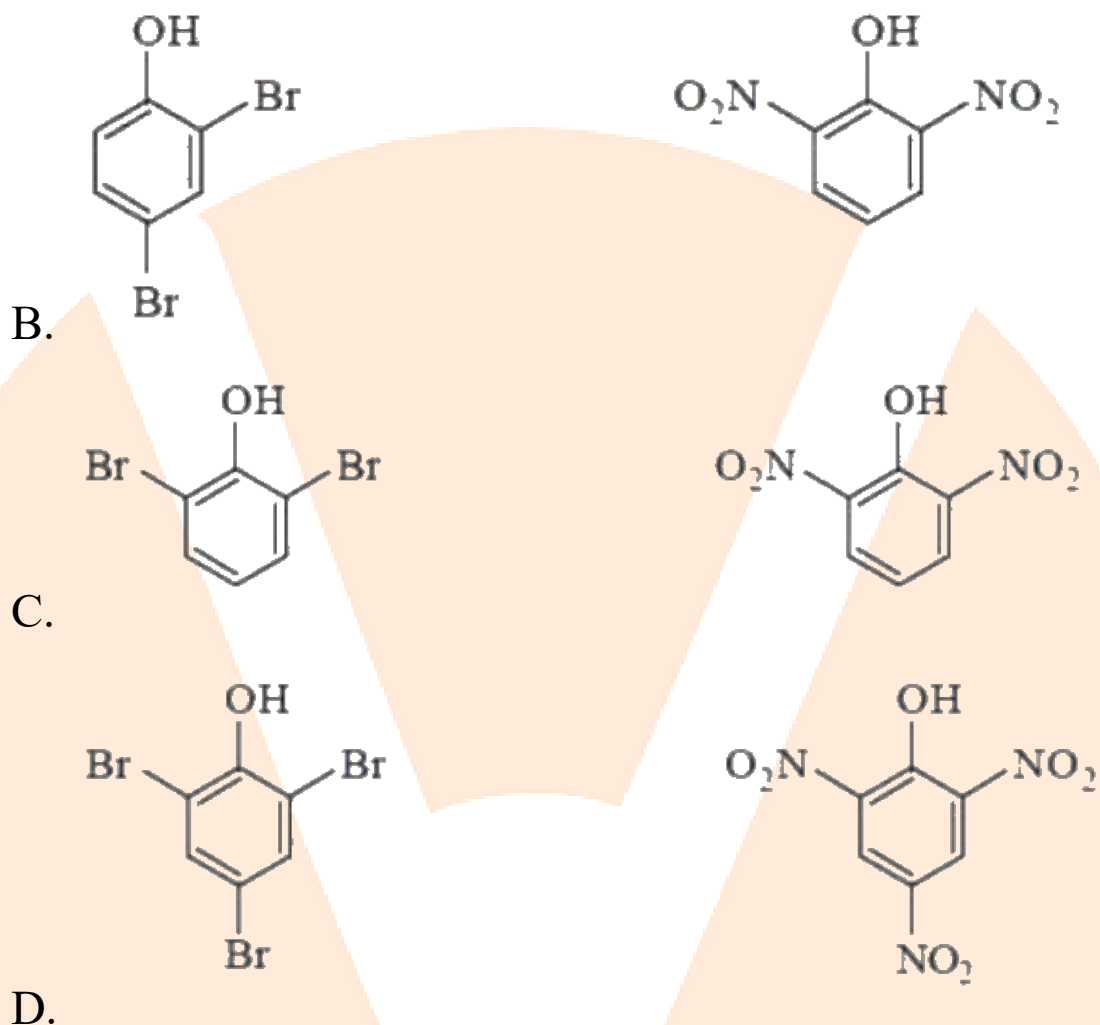
X



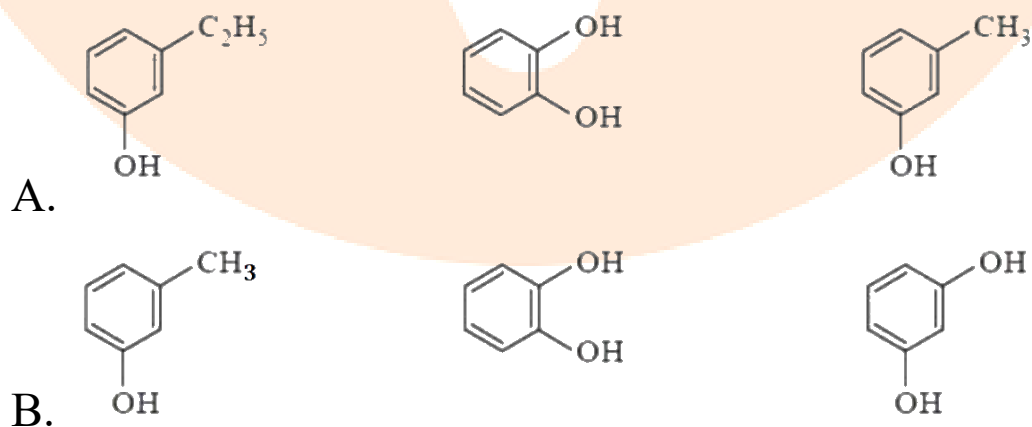
A.

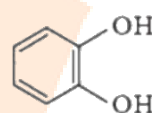
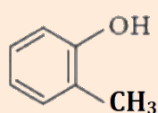
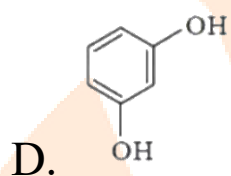
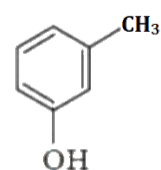
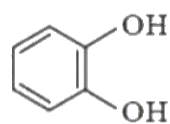
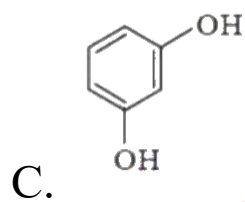
Y



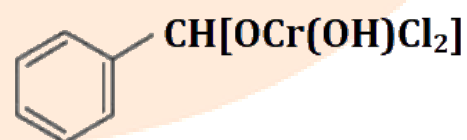
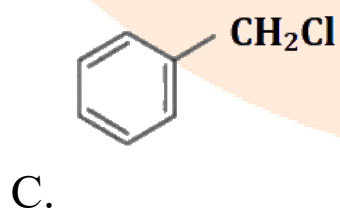
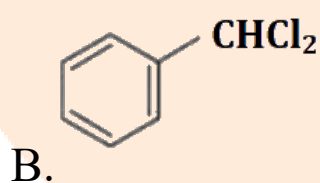
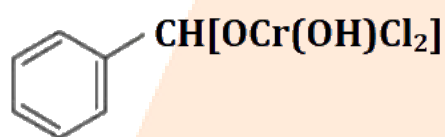
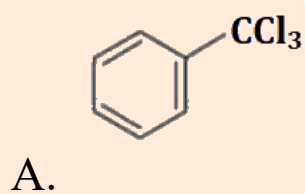
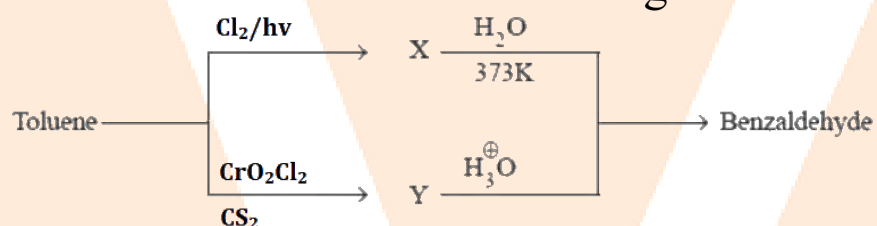


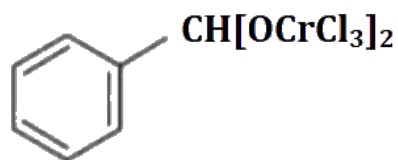
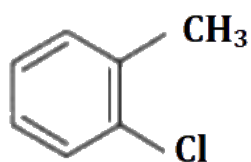
37. Identify m-cresol (A). catechol (B) and resorcinol (C) from the following



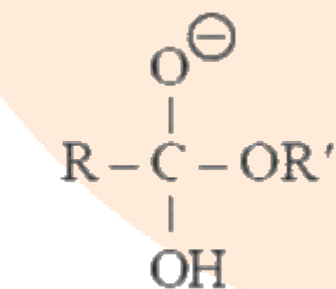
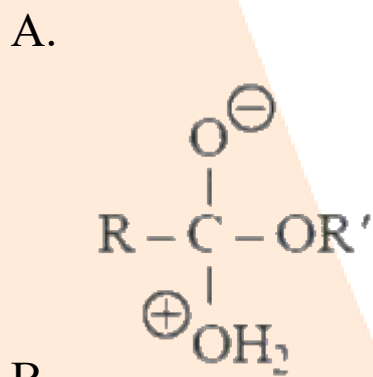
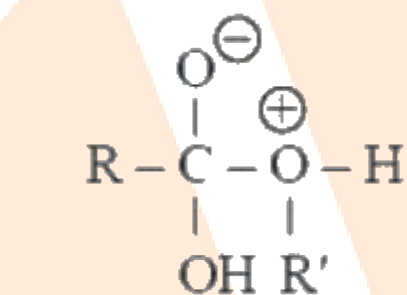


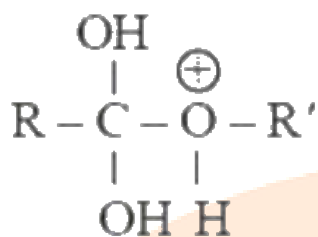
38. What are X and Y in the following reactions ?





39. Reaction of RCOOH with R'OH takes place in the presence of H_2SO_4 to form an ester. The intermediate in the reaction is



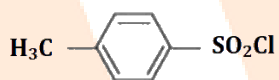


D.

40. The reagent used for the distinction of 1o, 2o and 3o

A. CH_3COCl

B. $\text{ZnCl}_2 \mid \text{HCl}$



D.

D. $[\text{Ag}(\text{NH}_3)_2]^+$